Solutions to JEE Advanced – 6 |JEE 2022 | Paper-2

PHYSICS

1.(BC) (a) $i = \frac{dq}{dt}$

$$i = u\int \rho\, dA = u\int_0^a cr^2 \times 2\pi r\, dr = 2\pi cu \times \frac{a^4}{4} = \frac{\pi cua^4}{2}$$

(b) $$\int EdA = E \times 2\pi rl = \frac{\int_0^r cr^2 \times 2\pi rldr}{\varepsilon_0} \Rightarrow E = \frac{c \times r^3}{4\varepsilon_0}$$

(c) $$B \times 2\pi r = \frac{\mu_0 \times \pi cur^4}{2} \Rightarrow B = \frac{\mu_0 cur^3}{4}$$

2.(ACD) Energy density $= \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0 \frac{Q_0^2}{(l^2\varepsilon_0)^2} = \frac{Q_0^2}{2l^4\varepsilon_0}$

$$U = \frac{1}{2}\varepsilon_0 E^2 \times volume = \frac{1}{2}\varepsilon_0\left(\frac{Q_0}{l^2\varepsilon_0}\right)^2 (l-x)ld = \frac{1}{2}\frac{Q_0^2}{l^3\varepsilon_0}(l-x)d$$

$$F = \frac{-dU}{dx} = -\left(\frac{1}{2}\frac{Q_0^2}{l^3\varepsilon_0}(-1)d\right) = \frac{Q_0^2 d}{2l^3\varepsilon_0}$$

3.(ABCD) Object distance, $u = -x$

Image distance, $v = \frac{uf}{u+f} = \frac{xf}{x-f}$

Distance between the object and image, $d = -u + v = x + \frac{xf}{x-f} = \frac{x^2}{x-f}$

4.(AD) $K_1 = eV$ and $K_2 = (2e)V$

$$\Rightarrow \frac{K_1}{K_2} = \frac{1}{2}$$

Now, $K_1 = \frac{1}{2}m_e v_1^2$ and $K_2 = \frac{1}{2}(4m_p)v_2^2$

$$\Rightarrow \frac{v_1}{v_2} = \left(\sqrt{\frac{4m_p}{m_e}}\right)\left(\sqrt{\frac{K_1}{K_2}}\right) = \sqrt{2}\left(\sqrt{\frac{m_p}{m_e}}\right) = \sqrt{3672} = 60.6$$

5.(ABD) Flux through the loop as a function of time,

$$\phi(t) = \int_a^{2a+vt}\left(\frac{\mu_0 I}{2\pi x}\right)(a)\,dx = \left(\frac{\mu_0 Ia}{2\pi}\right)\log_e\left(\frac{2a+vt}{a}\right)$$

$$\Rightarrow \phi\left(t = \frac{a}{v}\right) = \left(\frac{\mu_0 Ia}{2\pi}\right)\log_e(3)$$

Current through the loop as a function of time,

$$i(t)=\frac{1}{R_0}\frac{d\phi}{dt}=\frac{\mu_0 I a v}{2\pi R_0(2a+vt)}$$

$$\Rightarrow\ i\left(t=\frac{a}{v}\right)=\frac{\mu_0 I v}{6\pi R_0}$$

Also, we can deduce using Lenz law that the induced current is anticlockwise.

And, we can see that the magnitude of induced current decreases with time.

6.(AD) For constructive interference, optical path difference $P=n\lambda$

$$\Rightarrow \quad \sqrt{x^2+h^2}\left(\frac{4}{3}-1\right)=n\lambda$$

$$\Rightarrow \quad x^2+h^2=9n^2\lambda^2 \quad \Rightarrow \quad x=\sqrt{9n^2\lambda^2-h^2}$$

For destructive interference, path difference $P=(2n-1)\frac{\lambda}{2}$

$$\Rightarrow \quad \sqrt{x^2+h^2}\left(\frac{4}{3}-1\right)=(2n-1)\frac{\lambda}{2} \quad \Rightarrow \quad x^2+h^2=\frac{9}{4}(2n-1)^2\lambda^2$$

7.(BD) **(A)** Potential at the surface, $V_0=\frac{Q}{4\pi\varepsilon_0 R}$

Potential at a point at a distance *r* from the centre,

$$V_r=\frac{Q}{8\pi\varepsilon_0 R^3}\left(3R^2-r^2\right)=\frac{11}{8}V_0=\frac{11}{8}\left(\frac{Q}{4\pi\varepsilon_0 R}\right) \quad \Rightarrow \quad r=\frac{R}{2}$$

(B) Electric field at *P*, $E=\frac{Q\left(\frac{R}{2}\right)}{4\pi\varepsilon_0 R^3}=\frac{V_0}{2R}$

(C) $V_0=\frac{Q}{4\pi\varepsilon_0 R}=\frac{\rho\left(\frac{4\pi R^3}{3}\right)}{4\pi\varepsilon_0 R} \quad \Rightarrow \quad \rho=\frac{3\varepsilon_0 V_0}{R}$

(D) Recall the graph between potential and distance from centre for a positively charged sphere. The potential is maximum at the centre of the sphere.

8.(BD) For first refraction by curved surface.

$$\frac{\mu}{v_1}-\frac{1}{\infty}=\frac{\mu-1}{+R} \quad \Rightarrow \quad v_1=\frac{\mu R}{\mu-1}$$

For second refraction by flat surface

$$u=v_1-R=\frac{R}{\mu-1}$$

$$\frac{1}{v_2}-\frac{\mu}{\left(\frac{R}{\mu-1}\right)}=\frac{1-\mu}{\infty} \quad \Rightarrow \quad v_2=\frac{R}{\mu(\mu-1)}$$

This is the distance of final image from B.

$$\Rightarrow \quad d=\frac{R}{\mu(\mu-1)}$$

Since $d>0,$ the final image is outside the hemisphere.

Now, if light is incident on the flat surface,
For first refraction.

$$\frac{\mu}{v_1}-\frac{1}{\infty}=\frac{\mu-1}{\infty} \qquad \Rightarrow \qquad v_1=-\infty$$

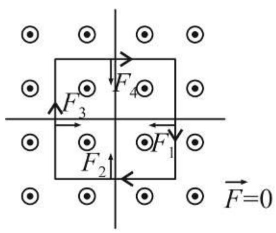
For second refraction

$$\frac{1}{v_2}-\frac{\mu}{-\infty}=\frac{1-\mu}{-R} \qquad \Rightarrow \qquad v_2=\frac{R}{\mu-1}$$

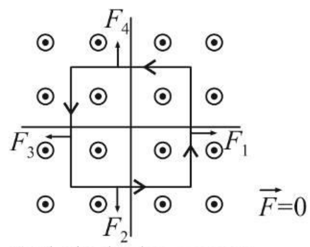
So, distance of this point from B $=\dfrac{R}{\mu-1}+R=\dfrac{\mu R}{\mu-1}$

9.(B), 10.(A)

If m is even :

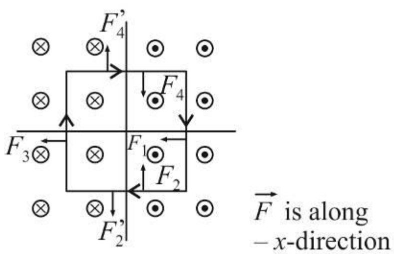


Clockwise current

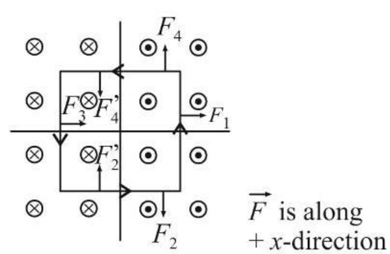


Anti clockwise current

If m is odd :



Clockwise current



Anti clockwise current

11.(C), 12.(B)

(p) $\qquad \lambda_{\min}=\dfrac{hc}{20\times10^3}=\dfrac{1240}{2\times10^4}=0.062$ nm

(q) $\qquad$ Kinetic energy of the photoelectron, $K=\dfrac{hc}{100}-\phi_0=\dfrac{1240}{100}-7.4=5.0\,\text{eV}$

De-Broglie wavelength of the photoelectron,

$$\lambda=\frac{h}{\sqrt{2mK}}=\frac{6.6\times10^{-34}}{\sqrt{2\left(9\times10^{-31}\right)\left(8\times10^{-19}\right)}}=0.55\,\text{nm}$$

(r) $\qquad \lambda=\dfrac{hc}{13.6(1)^2\left(\dfrac{1}{2^2}-\dfrac{1}{4^2}\right)}=\dfrac{1240}{(13.6)\left(\dfrac{3}{16}\right)}=486.3$ nm

(s) $\qquad$ Kinetic energy of the electron in the 4$^{\text{th}}$ Bohr orbit in a Hydrogen atom,

$$K=-E=13.6\left(\frac{1^2}{4^2}\right)=\frac{13.6}{16}\text{ eV}=\left(\frac{13.6}{16}\right)\left(1.6\times10^{-19}\right)=1.36\times10^{-19}\text{ J}$$

De-Broglie wavelength of the electron,

$$\lambda = \frac{h}{\sqrt{2mK}} = \frac{6.6\times10^{-34}}{\sqrt{2\left(9\times10^{-31}\right)\left(1.36\times10^{-19}\right)}} = \frac{6.6\times10^{-34}}{5\times10^{-25}} = 1.32\,\text{nm}$$

1.(0.25) Adding an identical rectangular surface with one vertex in the third quadrant and another vertex in the fourth quadrant, and then completing the cube around the charge at $(0, 0, a)$, we can see that the flux through the original rectangle due to $+Q$ is $\phi_1 = \frac{Q}{12\varepsilon_0}$ (in the $-z$ direction)

Similarly, the flux through original rectangle due to $-2Q$ is $\phi_2 = \frac{2Q}{12\varepsilon_0} = \frac{Q}{6\varepsilon_0}$ (in the $-z$ direction)

So, net flux: $\phi = \phi_1 + \phi_2 = \frac{Q}{4\varepsilon_0}$

2.(2) For the first capacitor, $\frac{1}{C_{eq}} = \frac{\left(\frac{d}{3}\right)}{K\varepsilon_0 A} + \frac{\left(\frac{2d}{3}\right)}{4K\varepsilon_0 A} = \frac{d}{2K\varepsilon_0 A} \Rightarrow C_{eq} = \frac{2K\varepsilon_0 A}{d}$

For the second capacitor, $C_{eq} = \frac{K_0\varepsilon_0 A}{d} \Rightarrow \frac{K_0}{K} = 2$

3.(80) $(V_C)_{rms} = I_{rms}\left(\frac{1}{\omega C}\right)$ and $(V_R)_{rms} = I_{rms}R$

$\Rightarrow \quad (V_R)_{rms} = (V_C)_{rms}(\omega RC) = (40)(200)(100)\left(10^{-4}\right) = 80V$

4.(3.41) Total energy released in the two step process,

$Q = \left(m\left({}^{131}_{53}I\right) - m\left({}^{131}_{54}Xe\right)\right)(931.5)\ MeV = 970.623\ keV$

Therefore, maximum energy that can be emitted as gamma radiation,

$E_{gamma} = Q - 606.623 = 364\ \text{keV}$

Wavelength of this gamma photon, $\lambda = \frac{1242}{364\times10^3} = 3.41\times10^{-12}m$

5.(8) Force $= \left[2(0.4)\frac{I}{C} + \left(0.2\frac{I}{C}\right) + 0\right]\times A \quad = \frac{I}{C}\times A = \frac{3000}{3\times10^8}\times100\times10^{-4} = 1\times10^{-7}$

$\therefore \quad x + y = 1 + 7 = 8$

6.(2) Let the current through the $5\,\Omega$ resistance be x in the direction shown. Then, by applying KCL we can assume the currents in the other branches as well. Now, starting from the positive terminal of the battery, applying KVL, going clockwise:

$-5x - 1(10) - 2(x+3) + 30 = 0 \quad \Rightarrow \quad x = 2\ \text{A}$

CHEMISTRY

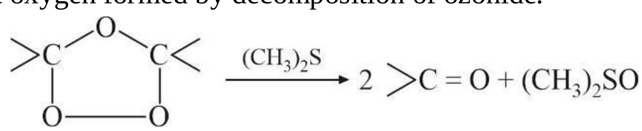
1.(ABCD) The main forces which stabilise the secondary and tertiary structures of proteins are hydrogen bonds, disulphide linkages, vander Waals and electrostatic forces of attraction.

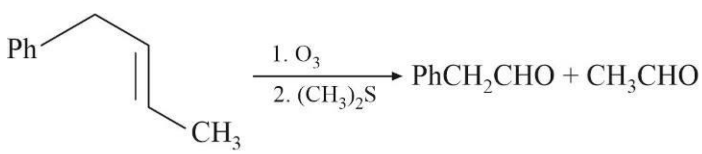
2.(ABCD)

- Colouration of halogen is due to HOMO (π^*) to LUMO (σ^*) charge transfer.
- Violet colouration of $KMnO_4$ is due to ligand to metal charge transfer i.e. from O to Mn.
  In this case d-d transition is not possible because of +7 oxidation state of Mn.
- According to crystal field theory, colouration of transition metal complex is due to d-d transition of unpaired electron from t_{2g} to e_g orbitals in case of octahedral complex ion.
- Solid ozone is violet black in colour hence it absorb visible radiations while in gas phase ozone absorb UV (255nm) radiations and appears pale blue in colour.

3.(ABC)

- Double bond in side chain easily undergoes ozonolysis. Ozonolysis in the presence of dimethylsulphide is reductive ozonolysis because $(CH_3)_2S$ acts as reducing agent and reacts with nascent oxygen formed by decomposition of ozonide.



- Acetaldehyde undergoes haloform reaction.
  $$CH_3CHO+3I_2+4NaOH \longrightarrow CHI_3+HCOONa+3NaI+3H_2O$$
- Priority of alkyl group is higher than the priority of H. In given compound substituent of higher priority are on opposite side of multiple bond $(C=C)$ hence its configuration is E not Z.
- 

  Mixture of $PhCH_2CHO$ and CH_3CHO in presence of NaOH undergoes aldol condensation reaction.

$$PhCH_2CHO+CH_3CHO \longrightarrow \underset{\text{(Self aldol)}}{PhCH_2CH=\overset{\overset{\displaystyle Ph}{|}}{C}-CHO} + \underset{\text{(Self aldol)}}{CH_3CH=CHCHO}$$

$$\underset{\text{(cross aldol)}}{PhCH_2CH=CH-CHO} + \underset{\substack{\text{(2-phehylbut-2-ene)}\\ \text{(cross aldol)}}}{CH_3-CH=\overset{\overset{\displaystyle Ph}{|}}{C}-CHO}$$

4.(ABCD)

Group I radicals are detected as insoluble chlorides by using dil. HCl.

- $PbCl_2$ is soluble in hot solution hence can't be precipitated in hot condition
- AgCl is soluble in aq. ammonia due to the formation of complex $Ag(NH_3)_2^+$ and can't be reprecipitated by addition of dil. HCl because coordinated ammonia can't reacts with HCl.
- $Hg_2Cl_2+2NH_3 \longrightarrow Hg(NH_2)Cl+Hg+NH_4Cl$ (disproportionation)

5.(ABD)

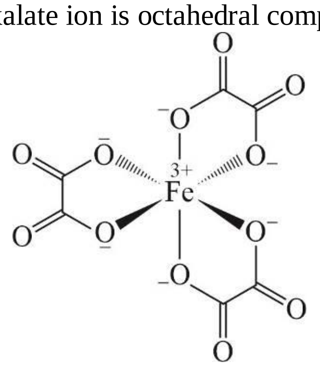
- When acidified $K_2Cr_2O_7$ is used as oxidising agent in a redox titration, an external indicator, diphenylamine is used.
- Soluble metal chlorides when heated with solid $K_2Cr_2O_7$ and conc. H_2SO_4 produce red coloured vapours of chromyl chloride. This reaction is used for identification of metal chloride and called as chromyl chloride test.
- Orange colouration of aq. $K_2Cr_2O_7$ is due to charge transfer from oxygen to chromium (not due to d-d transition because of absence of d-electrons in Cr of $K_2Cr_2O_7$).
- In alkaline medium dichromate ion hydrolyze to form yellow coloured chromate ion.

$$H_2O + Cr_2O_7^{2-} \rightleftharpoons 2CrO_4^{2-} + 2H^+$$

Alkaline medium shift this equilibrium towards forward direction.

6.(ABCD)

Ferrioxalate ion is octahedral complex ion

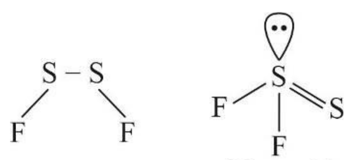


- This complex ion is completely dissymmetric hence chiral
- It has two different carbon-oxygen bond distances i.e. $C-O$ and $C=O$ bond distances
- Figure indicate six iron-oxygen bonds
- Oxalate acts as reducing agents and acidified MnO_4^- acts as strong oxidising agent

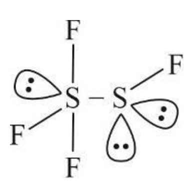
$$5Fe(C_2O_4)_3^{3-} + 6MnO_4^- + 48H^+ \longrightarrow 5Fe^{3+} + 30CO_2 + 6Mn^{2+} + 24H_2O$$

Five moles of ferrioxalate ions are oxidised by six moles of MnO_4^- ions.

7.(ABCD)

(A) 

More stable

(B) 

All fluorine have different environment, one of the F is bonded to tetrahedral sulphur. Both axial F atoms of trigonal bipyramidal sulphur are not in same environmental due to the other sulphur atom.

(C) SF_4 is polar covalent hence stronger intermolecular forces and less volatile while SF_6 is non-polar covalent and more volatile.

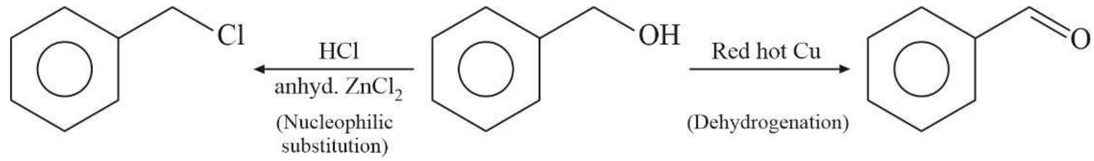
(D) SF_6 is exceptionally stable for steric reasons. The main contribution to the inertness of SF_6 is the steric hindrance of the sulphur atom, whereas its heavier group-16 counter parts, such as SeF_6 are more reactive than SF_6 as a result of less steric hindrance.

8.(ABC) For Rhombohedral crystal system unit cell, $\alpha = \beta = \gamma \neq 90°$. While for cubic, orthorhombic and tetragonal, $\alpha = \beta = \gamma = 90°$.

Sol. 9-10

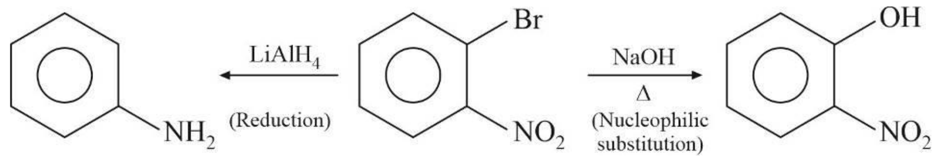
9.(B) 10.(B)

(P)



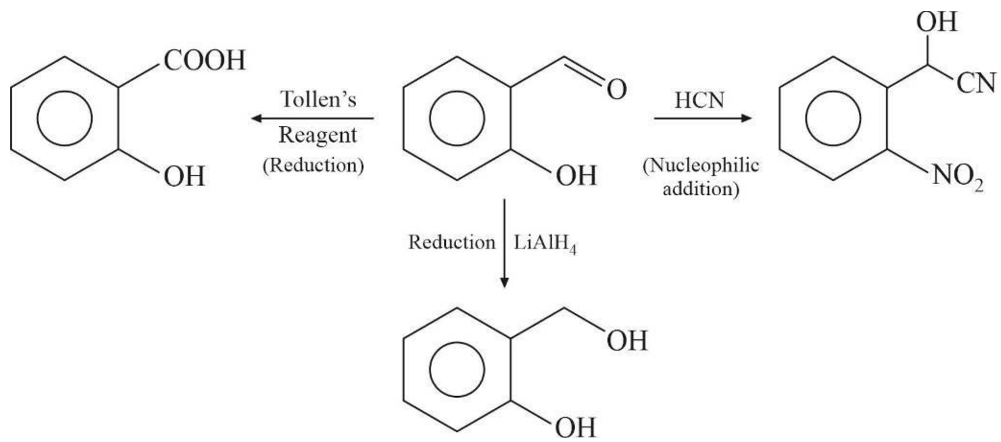
Nucleophilic addition, dehydrohalogenation, oxidation by Tollen's reagent and reduction by $LiAlH_4$ are not possible in benzyl alcohol.

(Q)



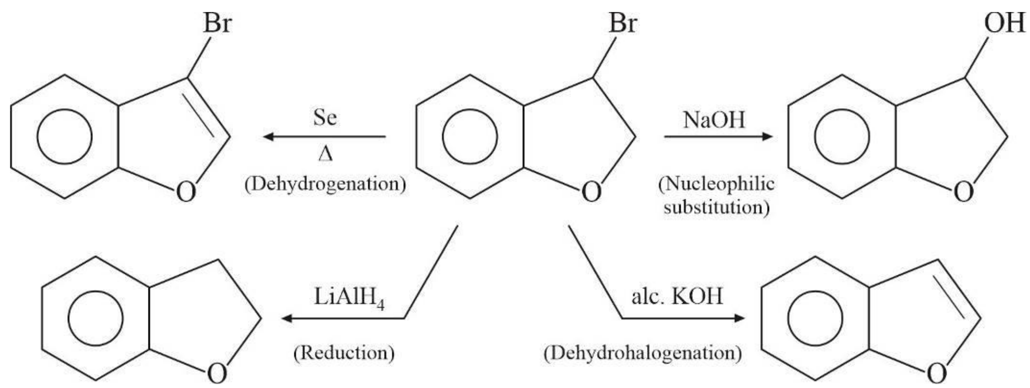
Dehydrogenation, nucleophilic addition, dehydrohalogenation and oxidation by Tollen's reagent are not possible in o-bromo nitrobenzene

(R)



Dehydrogenation, nucleophilic substitution and dehydrohalogenation are not possible in salicyldehyde

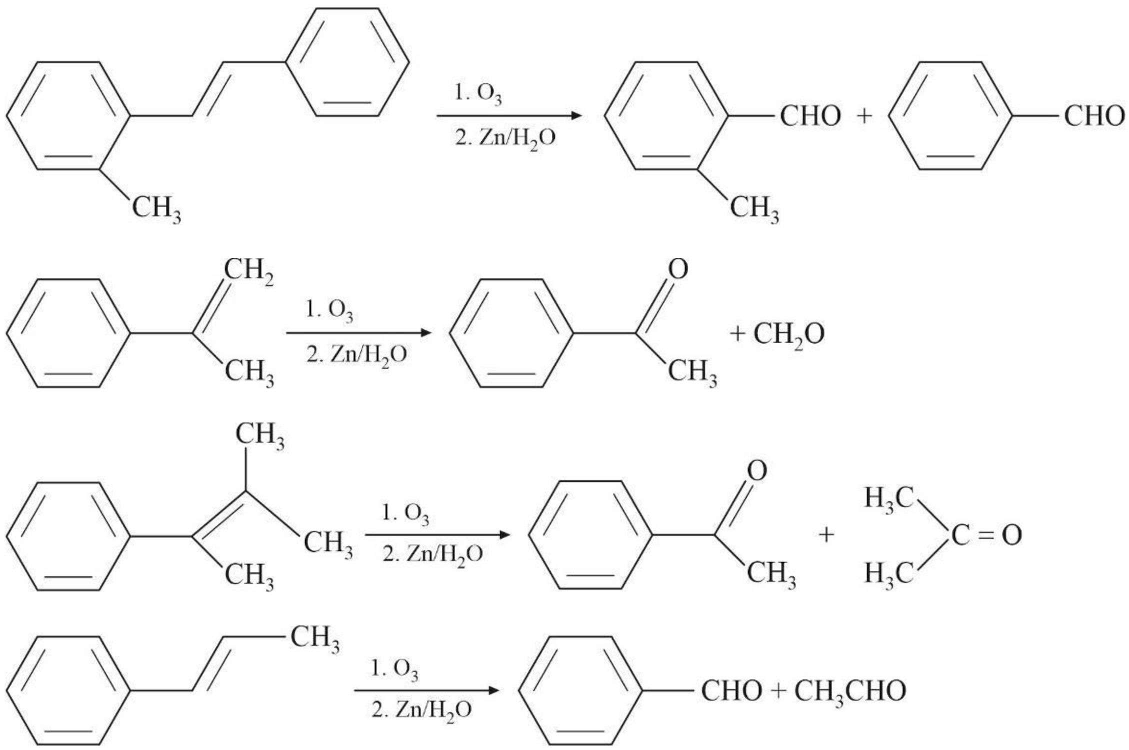
(S)



Nucleophilic addition and oxidation by Tollen's reagent are not possible in this compound

Sol. 11-12

11.(A) 12.(C)



- Aldehyde without alpha hydrogen undergoes cannizzaro reaction
- Methyl ketone and acetaldehyde undergoes haloform reaction
- All aldehydes form silver mirror with Tollen's reagent
- Carbonyl compounds having alpha hydrogen atom undergoes aldol condensation reaction
- All aldehyde except aromatic aldehyde reacts with Fehling's solutions

1.(88) Only ethanal reacts with Fehling's solution to form acetic acid and cuprous oxide (red ppt.)

$$CH_3CHO + 2Cu^{2+} + 2H_2O \longrightarrow CH_3COOH + Cu_2O + 4H^+$$

Mole of $CH_3CHO =$ Mole of Cu_2O

$$\frac{W}{44} = \frac{1.43}{143}$$

Percentage of ethanal $= \dfrac{1.43}{143} \times \dfrac{44}{0.5} \times 100 = 88$

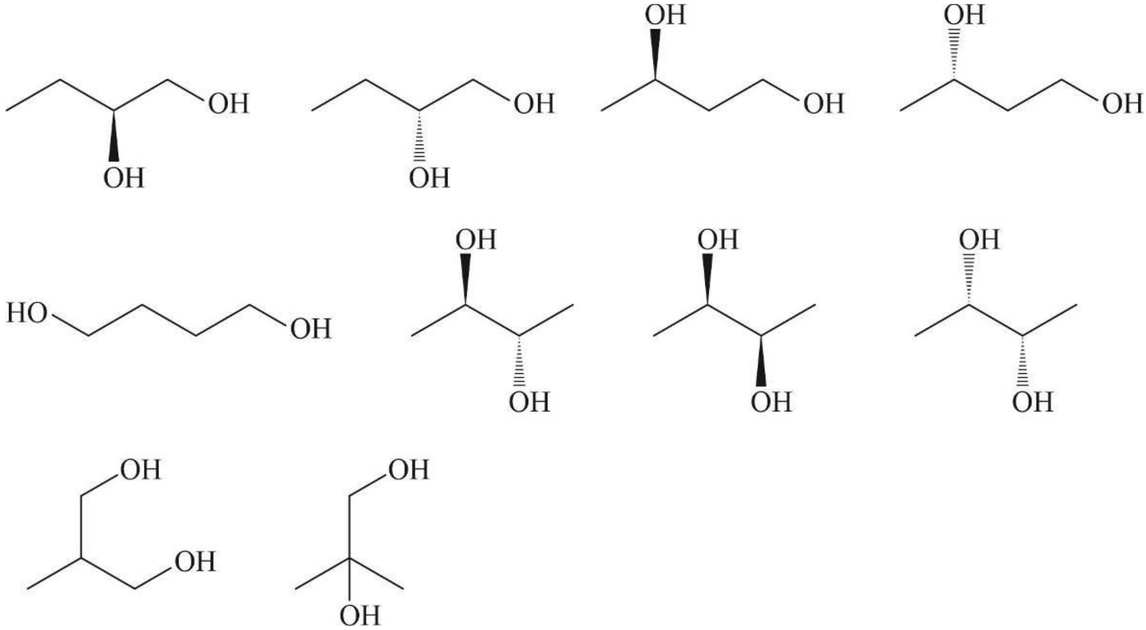
2.(10) Degree of unsaturation in compound $C_4H_{10}O_2$ is zero hence it can't be carboxylic acid. It is a hydroxy compound. Let assume that it contains n hydroxy groups.

$$2Z(OH)_n + 2nNa \longrightarrow 2Z(ONa)_n + nH_2$$

Mole of H_2 formed $= \dfrac{n}{2} \times$ mole of polyhydroxy compound

$$\frac{224}{22400} = \frac{n}{2} \times \frac{0.90}{90} \quad \Rightarrow \quad n = 2$$

Hence given compound is dihydroxy compound. Its possible isomers are given below.



3.(20000) Let be assume that there are n units of amino acid lysine in pepsin along with other amino acids.

Mole of lysine $= n \times$ mole of pepsin

$$\frac{0.73}{146} = n \times \frac{100}{M_0}$$

Molecular mass of protein, $M_0 = n \times \frac{100 \times 146}{0.73} = \frac{146 \times 10000}{73} \times n = 20000n$

For minimum molecular mass, $n = 1$

4.(62.50) For maximum theoretical percentage yield of M, percentage yields of N will be zero.

Maximum yield of $M = \frac{[M]}{[\text{all products}]} \times 1000 = \frac{5}{(5+3)} \times 100 = 62.5$

5.(147.78) $\pi V = \frac{w}{M_0} RT$

$$M_0 = \frac{w}{V} \times \frac{RT}{\pi} = 1.2 \times \frac{0.0821 \times 300}{0.20} = 147.78 \, g/mole$$

6.(0.34) In diamond cubic structure, all FCC lattice points and alternate tetrahedral voids are occupied. Effective number of constituent particles are 8.

Edge length of unit cell is related with radius by relation, $\frac{\sqrt{3}a}{4} = 2r$ because tetrahedral voids are occupied and $\frac{r_{void}}{r_{lattice}} > 0.225$.

$$\text{Atomic packing factor} = \frac{\text{volume occupied by all constituents}}{\text{volume of unit cell}} = \frac{8 \times \frac{4}{3}\pi r^3}{a^3} = \frac{8 \times \frac{4}{3}\pi r^3}{\left(\frac{8r}{\sqrt{3}}\right)^3} = \frac{\sqrt{3}\pi}{16} = 0.3399 \approx 0.34$$

MATHEMATICS

1.(ABCD) $t=\dfrac{1}{n};$ as $n\to\infty, t\to 0$

$$f(x)=\lim_{t\to 0}\left[\frac{x^{2017t}-x^{2018t}}{t}\right]=(2017)\ln x-(2018)\ln x \quad\Rightarrow\quad f(x)=-\ln x$$

2.(ABD) $Tr(A)=\sum_{i=1}^{3}\tan^{-1}(\tan 2i)=\tan^{-1}(\tan 2)+\tan^{-1}(\tan 4)+\tan^{-1}(\tan 6)$

$=2-\pi+4-\pi+6-2\pi=12-4\pi$

3.(BC) $f(x)=\sqrt{1-x^2}\left(\dfrac{1}{1+x^2}-\dfrac{x^2\cdot x^2}{1+x^2}\right)=(1-x^2)^{3/2}$

$$\int_0^1 f(x)\,dx=\int_0^1(1-x^2)^{3/2}\,dx=\int_0^{\pi/2}\cos^4\theta\,d\theta=\frac{3\pi}{16}$$

4.(B) Use IVT

Let $h(x)=x^{11}-f(x)$

$h(0)<0$ and $h(1)>0$

5.(BC) $$\lim_{x\to 0}\frac{\sqrt{1+x^2}\tan\left(\dfrac{x}{\sqrt{1+x^2}}\right)+2\sqrt{1-x^2}\sin\dfrac{x}{\sqrt{1-x^2}}-3x}{x^p}$$

Using expansion $$\lim_{x\to 0}\frac{\sqrt{1+x^2}\left(\dfrac{x}{\sqrt{1+x^2}}+\dfrac{x^3}{3(1+x^2)^{\frac{3}{2}}}+\dfrac{2}{15}\dfrac{x^5}{(1+x^2)^{\frac{5}{2}}}+\ldots..\right)+2\sqrt{1-x^2}\left(\dfrac{x}{\sqrt{1-x^2}}-\dfrac{x^3}{3!(1-x^2)^{\frac{3}{2}}}+\dfrac{x^5}{5!(1-x^2)^{\frac{5}{2}}}\ldots..\right)-3x}{x^p}$$

$$\lim_{x\to 0}\frac{\left(-\dfrac{2}{3}+\dfrac{2}{15}\dfrac{1}{(1+x^2)^2}+\dfrac{2}{5!(1-x^2)^2}\right)x^5}{x^p}\quad\Rightarrow\quad p=5\text{ and limit }=-\frac{2}{3}+\frac{2}{15}+\frac{2}{5!}=\frac{-31}{60}$$

6.(ABD) All the values of $|A|=\pm 4,\,0$ and determinant is 0 when two rows/columns are proportional

7.(AC) $$L=\lim_{n\to\infty}\frac{n^3\sum_{r=1}^{n}e^{r/n}}{(n+1)^m\sum_{r=1}^{n}r^{2m}}=\lim_{n\to\infty}\frac{n^4\sum_{r=1}^{n}e^{r/n}\cdot\frac{1}{n}}{n^{2m+1}(n+1)^m\sum_{r=1}^{n}\left(\frac{r}{n}\right)^{2m}\cdot\frac{1}{n}}$$

For L to be non-zero finite $m=1$

$$L=\lim_{n\to\infty}\frac{\frac{1}{n}\sum_{r=1}^{n}e^{r/n}}{\frac{1}{n}\sum_{r=1}^{n}\left(\frac{r}{n}\right)^{2m}}=\frac{\int_0^1 e^x\,dx}{\int_0^1 x^2\,dx}=\frac{e-1}{1/3}=3(e-1)$$

8.(BC) Let $P(x)=f(x)-x^2=$ four degree polynomial

$\because \quad P(1)=P(2)=P(3)=0$

$\Rightarrow \quad P(x)=(x-1)(x-2)(x-3)\ (x-\alpha) \Rightarrow \quad f(x)=x^2+(x-\alpha)(x-1)(x-2)(x-3)$

$\Rightarrow \quad f'(1)=2+(1-\alpha)(1-2)(1-3)=0$

$\Rightarrow \quad 2+2-2\alpha=0$

$\alpha=2$

$\Rightarrow \quad f(x)=x^2+(x-1)(x-2)^2(x-3)$

Now

$$\because \quad \begin{cases} P(1)=P(2)\Rightarrow P'(c_1)=0 \text{ for some } c_1\in(1,2) \\ P(2)=P(3)\Rightarrow P'(c_2)=0 \text{ for some } c_2\in(2,3) \end{cases}$$

$\Rightarrow \quad P''(c_3)=0$ for some $c_3\in(c_1,c_2)\subset(1,3) \quad \Rightarrow \quad f''(x)=2$ for some $x\in(1,3)\subset(0,4)$

9.(C)-10.(A)

9-10. (P) $|A_{2\times1}B_{1\times2}|$ is 0 (Q) $|A|=0$ and $|B|=0$

(R) $AB^T=B^TA$ (S) $|2A^2B^{-3}|=8\times1^2\times\frac{1}{27}$

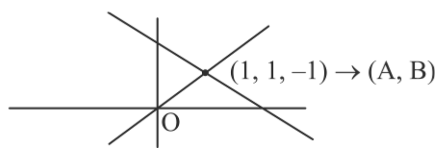
11.(A)-12.(B)

11-12. **(P)** $d=2\sqrt{3}$

(Q) These lines are skew and O lies on shortest distance

(R) Lines are parallel and O lies mid way between them

(S) Lines are coplanar and perpendicular



1.(2) $$\begin{vmatrix} 1-\lambda & 1 & 0 \\ -1 & -\lambda & 1 \\ 2 & 1 & -\lambda \end{vmatrix}=0$$

$(1-\lambda)(\lambda^2-1)-(\lambda-2)=0$

$-\lambda^3+\lambda^2+1=0$

$A^3-A^2-I=0$

$A^3=A^2+I$

2.(4) When x is $\frac{\pi}{4}^-$ then $\sqrt{2}\cos x+2>3$

3.(2.25) $4f^3(x)=f^3(2x+1)+c$

By taking $x=-\frac{1}{8},\frac{3}{4},\frac{5}{2}$ and 6, we get

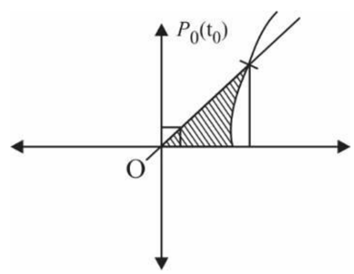
$f^3\left(\frac{3}{4}\right)+c=4,\ f^3\left(\frac{5}{2}\right)+c=4f^3\left(\frac{3}{4}\right),\ 4f^3\left(\frac{5}{2}\right)=f^3(6)+c$

So, $c=\frac{8}{3}$ i.e., $4f^3(x)=f^3(2x+1)+\frac{8}{3}$

4.(480) $\frac{1}{2}\left(\frac{e^{t_0}+e^{-t_0}}{2}\right)\left(\frac{e^{t_0}-e^{-t_0}}{2}\right)-\int_0^{t_0} y\,dx$

$$\frac{1}{8}\left(e^{2t_0}-e^{-2t_0}\right)-\int_0^{t_0}\frac{(e^t-e^{-t})^2}{4}dt=\frac{t_0}{2}=240$$

$$t_0=480$$



5.(21) $\begin{vmatrix} a & d & 1 \\ b & e & 1 \\ c & f & 1 \end{vmatrix}=-5, \begin{vmatrix} a & d & 1 \\ b & e & 2 \\ c & f & 3 \end{vmatrix}=3$

$$2\begin{vmatrix} a & d & 1 \\ b & e & 2 \\ c & f & 3 \end{vmatrix}-3\begin{vmatrix} a & d & 1 \\ b & e & 1 \\ c & f & 1 \end{vmatrix}=21$$

$$\begin{vmatrix} a & d & -1 \\ b & e & 1 \\ c & f & 3 \end{vmatrix}=21$$

6.(61.25) $\frac{dx}{dy}e^{-x}-\frac{1}{y}e^{-x}=\frac{1}{y^3}$

$$\frac{d}{dy}(e^{-x}y)=-\frac{1}{y^2},\ e^{-x}y=\frac{1}{y}+c$$